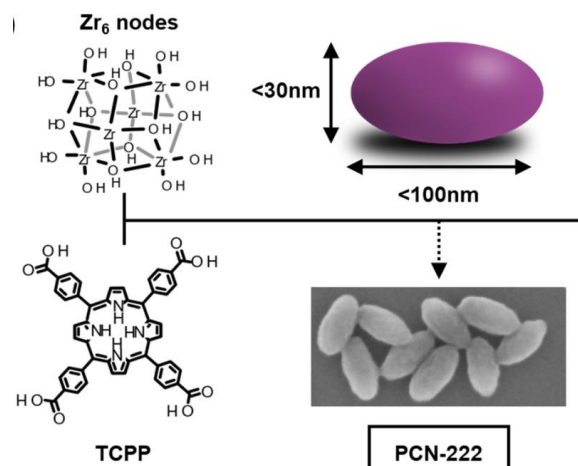


Zr₆ Nodes / Clusters

These nodes are fundamental to produce the MOF PCN-222¹ having specific size with good size distribution (they can be used also for other types of MOFs):



The procedure is published in previous work,² and described in our works,³ but here is reported with more details and tips. Even though the procedure is quite simple, it is crucial to check that the crystal are formed AND that there is no residual benzoic acid (since benzoic acid is a modulator for the preparation of MOFs, and it competes with the selected acid modulator, for example acetic acid for PCN222).

¹ X. Chen, Y. Zhuang, N. Rampal, R. Hewitt, G. Divitini, C. A. O'Keefe, X. Liu, D. J. Whitaker, J. W. Wills, R. Jugdaohsingh, J. J. Powell, H. Yu, C. P. Grey, O. A. Scherman and D. Fairen-Jimenez, J. Am. Chem. Soc., 2021, 143, 13557–13572.

² H. Noh, C.-W. Kung, T. Islamoglu, A. W. Peters, Y. Liao, P. Li, S. J. Garibay, X. Zhang, M. R. DeStefano, J. T. Hupp and O. K. Farha, Chem Mater, 2018, 30, 2193–2197. N. Planas, J. E. Mondloch, S. Tussupbayev, J. Borycz, L. Gagliardi, J. T. Hupp, O. K. Farha and C. J. Cramer, J Phys Chem Lett, 2014, 5, 3716–3723.

³ 10.1039/d3gc02774k; 10.1039/d4gc03743j

Typical reaction (adapted for our lab)

Materials

80 wt% $\text{Zr}(\text{OBu})_4$ (**Zirconium(IV) buotoxide**) in **n-butanol**: 7.5 mL (15.8 mmol, MW 383.67 g mol⁻¹). **NOTE**: if the bottle is full of white precipitates, please use a new one.

- **Benzoic Acid**: 50 g (0.82 mol, MW 122.12 g mol⁻¹)
- **Anhydrous Isopropanol (iPrOH)** or **n-propanol**: 150 mL. **NOTE**: the "classical" synthesis utilizes n-propanol (b.p. 97 °C). However, we prefer isopropanol (b.p. 82 °C) as a greener, bio-derived alternative. The main difference is in the reaction kinetics (see reflux temperature): the white cluster begins to form a visible white precipitate sooner with iPrOH. In n-propanol, the reaction mixture remains clear for longer before precipitation occurs because n-propanol solvates the intermediate species more effectively.

Procedure

1. Weigh 50 g of benzoic acid into a round-bottom flask (250mL) equipped with a reflux condenser and a magnetic stir bar, then purge the system with dry Argon (or Nitrogen).
2. Under a continuous flow of Ar/nitrogen, transfer 150 mL of anhydrous iPrOH (or n-propanol) into the flask. Start stirring the solution until complete dissolution of the benzoic acid, using also a US bath. **NOTE**: it takes 15/20' to completely dissolve the benzoic acid in the US bath.
3. Add 7.5 mL of 80 wt% $\text{Zr}(\text{OBu})_4$. **NOTE**: the reagent is quite viscous. Since this is quite impossible to add it under perfect inert atmosphere, add the reagent as quickly as possible (in the middle of the stirring alcohol, still not in once, but making drops) using a 10 mL plastic syringe without a needle, opening and closing the flask neck rapidly (place the syringe and close with some parafilm during the addition) to minimize atmospheric moisture exposure.
4. Heat the mixture to reflux (please remember to open the water circulation in the reflux) under constant magnetic stirring and an active Ar/nitrogen atmosphere overnight. **NOTE**: at least 12h, no need to get crazy, technically the nodes are formed within a few hours, but it is better to prepare them with no hurry.
5. Turn off heating and allow the mixture to cool. Remove the excess solvent by vacuum distillation (rotavapor) with mild heating (40 °C more or less) to yield a white solid powder.
6. Wash the resulting white solid **extensively** with cold iPrOH via vacuum filtration to **remove unreacted benzoic acid**. **NOTE**: it takes time!!! (2/3 h).
7. Dry the powder under high vacuum at room temperature (in line).

Characterization

To ensure high purity of the Zr_6 cluster and prevent stoichiometric or structural errors during PCN-222 preparation (**I repeat again: there must not be residual benzoic acid, thanks! ☺**), verification via FTIR and XRD is fundamental.

ATR-FTIR

Record ATR-FTIR spectrum (old building) of white powder to confirm the complete removal of unreacted benzoic acid:

- **Free Benzoic Acid Indicator:** Look specifically for a sharp, intense band at $1680\text{--}1700\text{ cm}^{-1}$, corresponding to the carbonyl ($C=O$) stretching vibration of free, protonated carboxylic acid.
- **Coordinated Benzoate Core:** Pure $[Zr_6(\mu_3-O)_4(\mu_3-OH)_4(PhCOO)_{12}]$ clusters display no peak at $1680\text{--}1700\text{ cm}^{-1}$. Instead, they exhibit characteristic symmetric (ν_s) and asymmetric (ν_{as}) carboxylate (COO^-) stretching frequencies shifted to $\approx 1415\text{ cm}^{-1}$ and $\approx 1540\text{ cm}^{-1}$, confirming full coordination to the Zr(IV) nodes.
- **Action Threshold:** If the $1680\text{--}1700\text{ cm}^{-1}$ peak persists, perform additional wash cycles with cold iPrOH followed by re-drying under high vacuum until the free acid signal disappears completely.

XRD

Collect a XRD pattern (old building) ($2\theta = 2^\circ\text{--}30^\circ$) (**NOTE: it would be better to measure to a smaller angle, but our equipment cannot**) to evaluate the crystallinity and structural phase of the synthesized cluster:

- **Cluster Diffraction Profile:** Pure discrete Zr_6 benzoate clusters exhibit sharp, distinct low-angle diffraction peaks corresponding to the molecular arrangement of the isolated cluster framework.
- **Absence of ZrO_2 Impurities:** Confirm the complete absence of broad halo humps or characteristic reflections of monoclinic/tetragonal zirconium oxide (ZrO_2 , typically visible around $2\theta \approx 28^\circ\text{--}32^\circ$), which would indicate premature hydrolysis from atmospheric moisture during setup.

Tips

- **Inert Atmosphere:** Zirconium alkoxides ($Zr(OBu)_4$) are sensitive to moisture and oxygen. Operating under an Argon atmosphere (or nitrogen) prevents premature hydrolysis, which otherwise leads to the formation of amorphous, insoluble zirconium

oxides or poorly defined hydroxides instead of the discrete, crystalline $[\text{Zr}_6\text{O}_4(\text{OH})_4(\text{PhCOO})_{12}]$ cluster core. To be honest, it is far more important to open and seal the bottle of the reagent under inert atmosphere, more than the synthesis itself (we don't want to waste a bottle every preparation of the nodes).

- **Benzoic Acid:** The excess benzoic acid coordinates directly to the forming Zr_6 octahedral core, capping the 12 coordination sites of the node with benzoate ligands. This pre-assembled, benzoate-capped node provides high stability and reproducible stoichiometry ($\text{MW } 1164 \text{ g mol}^{-1}$ per node unit) for the subsequent MOF assembly.